Full application

In progress

Early-Career Award

Application title

Quantifying Care Pathway Delays and Modelling Mental Health Symptom Trajectories among Adolescents in Kenyan Schools: A Methodological and Public Health Program

Lead applicant

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Application reference

345293/Z/26/Z

Administering organisation

Kenya Institute of Primate Research

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Application summary

Application title

Quantifying Care Pathway Delays and Modelling Mental Health Symptom Trajectories among Adolescents in Kenyan Schools: A Methodological and Public Health Program

Proposed length of funding (months)

60

Proposed start date

10/10/2026

Research subject area

Data Sciences, Tools and Technology

Administering organisation

Kenya Institute of Primate Research
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Completed

Research proposal summary

Research summary

This programme confronts systemic barriers that delay identification and support for adolescents with ADHD in Kenyan schools. Moving beyond prevalence snapshots, it investigates “invisible attrition” - the structural, institutional, and social frictions that prevent timely recognition and exacerbate educational disadvantage. Focusing on the transition from upper primary to junior secondary school, I will reconstruct each learner’s pathway from first concern to formal referral, mapping delays across teachers, parents, and clinical services. Cerebral palsy serves as a visible comparator, enabling quantification of excess delays for less overt neurodevelopmental conditions via a stigma-delay coefficient. Using joint longitudinal–survival modelling, symptom trajectories will be linked to grade repetition, school transfer, and dropout. Multi-informant assessments and inverse-probability weighting ensure early exits are incorporated, capturing the experiences of the most vulnerable learners. A locally calibrated decision-support tool will translate these insights into probabilistic guidance for school-based interventions, aligning with national mental health policy and providing a scalable pathway to reduce invisible attrition and strengthen educational equity across sub-Saharan Africa. Keywords: ADHD, neurodevelopment, invisible attrition, recognition delay, Bayesian modelling, decision-support, Kenya.

Is this or a similar application for funding currently under consideration elsewhere?

No

Is this a resubmission of an Early-Career Award application submitted to Wellcome?

No

Completed

Your research vision

Research vision

Research vision, ambition and conceptual framework

The crisis of invisible attrition in Kenyan schools.

My research vision is to transform the identification of neurodevelopmental disorders in SSA (sub-Saharan Africa), shifting the focus from prevalence snapshots to quantifying systemic friction in care pathways. In Kenya, where nearly 90% of school aged children attend school, the transition from upper primary (Grades 4–6) to JSS (Junior Secondary School) (Grades 7–9) under the CBC (Competency-Based Curriculum) represents a period of developmental and institutional volatility. Academic expectations intensify, executive-function demands rise, peer hierarchies reorganize, and instructional styles shift toward greater learner autonomy.

This transition is consequential for a spectrum of neurodevelopmental and emotional disorders, including ADHD (Attention Deficit Hyperactivity Disorders), specific learning disorders, autism spectrum disorder, anxiety, and depression. Many remain under-identified, particularly where specialist assessment pathways are limited such as in SSA countries. Among them, ADHD warrants focused investigation.

Core impairments in inattention, impulsivity, and executive dysfunction interact directly with the organizational and planning demands of the CBC transition. Consequently, JSS may amplify previously compensated vulnerabilities, leading to grade repetition, behavioral referral, or school exit.

Kenyan studies suggest roughly one in five learners (21%) exhibit clinically significant ADHD symptoms , yet formal diagnosis within public schools is rare, leaving up to 60-80% of cases undetected. Untreated attentional and executive-function difficulties are often misinterpreted as indiscipline or low effort, initiating trajectories of educational exclusion, a phenomenon I term "invisible attrition".

Globally, ADHD research in Africa remains largely cross-sectional , providing prevalence snapshots but rarely interrogating timing, mechanisms, or structural bottlenecks that determine whether a child receives support. My programme directly addresses this structural void.

Conceptual shift: Informant discrepancy as inferential signal

A key barrier to ADHD identification in Kenya is low concordance between informants. Teachers observe executive functioning within structured classrooms, while parents observe emotional regulation at home. Meta-analytic evidence shows correlations between teacher and parent ratings are low to moderate, often 0.20–0.40. These discrepancies are commonly treated as measurement error. I propose a different perspective: informant discrepancy reflects meaningful context-specific signal.

In low-resource settings, diagnostic pathways are non-linear. Children may not progress reliably from parental concern to specialist assessment; recognition is fragmented by institutional, cultural, and social filters. By applying a Bayesian misclassification framework anchored in a clinically validated subsample, I will model latent disorder status underlying discrepant reports, weighting informant inputs by validation evidence to generate probabilistic estimates of true ADHD status.

This approach has two scientific advantages. First, it improves identification of inattentive presentations, particularly among girls, who are under-recognized in hyperactivity-focused screening systems. Second, it transforms disagreement into structured information about context-specific manifestation, preserving heterogeneity to refine posterior probabilities rather than collapsing multi-informant data into single scores.

Quantifying systemic friction and recognition delay.

Beyond identifying symptoms, this programme examines how long it takes for concerns about a child's behavior to translate into formal recognition within the education and health systems. For each learner, I will reconstruct the pathway from the first sustained concern reported by a parent or teacher to documented referral for specialist assessment at an EARC or clinical service. Where exact dates are uncertain, timing will be recorded within defined intervals to minimize recall bias and analyzed using interval-based methods.

Rather than relying on average waiting times, the full distribution of recognition delays will be examined to determine how the likelihood of referral evolves over time. The pathway will be decomposed into sequential stages - initial awareness, teacher recognition, parental action, and clinical referral - and this will allow identification of where delays accumulate. This stage-specific analysis makes it possible to distinguish school-level, household-level, and health-system bottlenecks. Differences by county, sex, and symptom presentation will reveal whether particular groups experience systematically prolonged recognition.

CP (Cerebral palsy) will serve as a comparator condition within the same counties. Because CP presents with visible motor impairment, recognition is typically less dependent on behavioral interpretation and classroom judgement. By comparing adjusted time-to-identification patterns between ADHD and CP, I will estimate the additional delay experienced by learners with ADHD after accounting for socioeconomic and geographic context. This excess delay - conceptualized as a stigma-delay coefficient - will quantify the structural disadvantage associated with less visible neurodevelopmental conditions and provides an interpretable metric for policy discussion.

Agreement between parent and teacher reports of ADHD symptoms is known to be modest, often around 0.30, reflecting contextual variation in behavior across home and school environments. To avoid anchoring delay estimates to a single perspective, onset timing will be treated probabilistically when informant reports diverge. In this way, informant discrepancy is incorporated into the estimation process rather than treated as measurement error.

Through this framework, recognition delay is measured not merely as waiting time, but as an observable indicator of how institutional interpretation, visibility of symptoms, and system design shape access to support. This moves the discussion from individual blame to structural response.

Addressing informative dropout: Hidden bias in longitudinal studies.

Building on the earlier reconstruction of recognition pathways, the programme next considers the consequences of delayed or missed identification. In low-resource settings, learners whose ADHD symptoms remain untreated are more likely to repeat grades, transfer schools, or leave the system entirely. These exits are not random: they are directly shaped by symptom burden, visibility of difficulties, and structural friction in care pathways.

To capture this informative dropout, I will apply joint longitudinal–survival modelling that links the trajectory of ADHD symptoms with event-time outcomes, including grade repetition, school transfer, and dropout. Repeated multi-informant assessments from parents, teachers, and clinical evaluations will be combined with timing of school events to estimate how the persistence, intensity, and slope of untreated symptoms drive educational attrition.

Because learners most affected by ADHD may leave the system before full observation, IPW (inverse-probability weighting) will be used to reconstruct population-level trajectories. This approach will ensure that analyses account for early exits, giving an accurate representation of the experiences of the most vulnerable adolescents.

By connecting structural delays in recognition with measurable outcomes, this approach makes explicit how the timing of identification cascades into educational disadvantage. It transforms multi-informant, longitudinal data into actionable insight, showing not just who is affected, but how and when systemic friction contributes to invisible attrition. This integrated perspective will strengthen both the scientific foundation and the translational potential of the programme.

Translational component: From inference to school-level decision support.

The ultimate aim of this programme is to transform statistical insight into practical action through a locally calibrated decision-support prototype. This tool will combine teacher and parent observations, multi-informant uncertainty, and longitudinal symptom trajectories to generate probabilistic estimates of latent ADHD status and projected educational risk. By integrating Bayesian misclassification with joint longitudinal–survival models, the system explicitly accounts for delayed recognition, symptom persistence, and structural friction in care pathways.

Outputs will be linked to context-appropriate, guideline-consistent next steps. These include school-based accommodations, referral to EARCs, or clinical evaluation in line with Kenya's Mental Health Policy (2015–2030). Within pilot counties, the tool's calibration, discrimination, and decision-curve performance will be rigorously evaluated, ensuring that probabilities translate into actionable, trustworthy guidance for teachers and policymakers.

The intellectual contribution extends beyond digitization. It lies in transforming uncertainty-aware inference into a decision architecture that respects local context, heterogeneous symptom expression, and resource constraints. This bridges longitudinal research and policy implementation, providing a model for evidence-informed intervention in low-resource educational settings and offering a scalable pathway to reduce invisible attrition among neurodivergent adolescents.

Central research question.

Why are many Kenyan adolescents with ADHD identified late, and how do delays in recognition contribute to worsening symptoms, grade repetition, and school dropout?

Specific aims.

Aim 1: Measure time-to-identification delays and map fail-points across teacher, parent, and clinical pathways.

Aim 2: Track ADHD symptom trajectories over three years and examine how untreated

symptom burden affects grade repetition or school exit.

Aim 3: Develop and internally validate a practical decision-support prototype combining teacher, parent, and clinical information to provide timely, evidence-informed referral recommendations.

Background, groundwork, and feasibility

This programme builds on existing collaborations with public primary and junior secondary schools, county-level educational officers, and EARCs in two Kenyan counties representing urban and peri-urban settings. Preliminary feasibility work has established the acceptability of teacher- and parent-reported screening tools and confirmed data linkage pathways between school records and referral documentation. My prior training in longitudinal modelling, missing data methods, and Bayesian inference provides the analytical foundation for implementing misclassification and joint modelling frameworks in complex, real-world data environments. These partnerships and methodological preparations ensure both operational feasibility and analytic rigour.

All data collection will adhere to national ethical guidelines, with institutional ethical approval, informed parental consent, adolescent assent, and secure data governance procedures aligned with Kenyan data protection laws (DP Act 2019) and standards.

Key stages, location, and milestones

The programme will be conducted in two Kenyan counties representing urban and peri-urban educational contexts, in partnership with public upper primary schools, junior secondary schools, and EARCs.

Year 1: Retrospective pathway reconstruction and validation groundwork

Year 1 will focus on retrospective mapping of recognition pathways. Structured interviews and record abstraction will reconstruct time-to-identification trajectories for ADHD and cerebral palsy. A clinically assessed validation subsample will be established to estimate sensitivity and specificity of teacher and parent reports. Preliminary delay-time models will be developed, and the stigma-delay coefficient will be estimated. Ethical approvals, data governance frameworks, and instrument adaptation will be finalized.

Year 2: Prospective cohort establishment and baseline modelling

A prospective cohort of learners transitioning from upper primary to JSS will be recruited. Baseline multi-informant assessments will be conducted, and linkage systems for grade progression, repetition, transfer, and referral outcomes will be operationalized. Bayesian misclassification models anchored in the Year 1 validation data will be implemented to

estimate latent ADHD status. Initial joint longitudinal–survival model specifications will be developed.

Year 3: Longitudinal follow-up and joint modelling refinement

Repeated assessments will be conducted to capture symptom trajectories over time. Joint longitudinal–survival models will be fitted to estimate associations between symptom slope, persistence, and risk of grade repetition or school exit. Competing-risk structures will be evaluated. Inverse-probability weighting strategies will be implemented to address informative dropout. Internal validation procedures, including bootstrap-based calibration and discrimination analysis, will be undertaken.

Year 4: Model validation, generalizability testing, and prototype development

Predictive models will undergo temporal validation within the cohort and geographic validation across participating schools. Model transportability across sex and county will be assessed. The decision-support prototype integrating Bayesian classification and risk prediction outputs will be developed in collaboration with educators and EARCs. Usability testing and refinement will be conducted through structured stakeholder workshops.

Year 5: Pilot implementation, policy integration, and capacity building

The prototype will be piloted within participating schools and EARCs to evaluate implementation feasibility, decision utility, and acceptability. Decision-curve analysis will quantify net benefit under realistic referral thresholds. Findings will be disseminated through county and national policy dialogues. Training workshops in longitudinal and Bayesian modelling will build local analytic capacity. Final outputs will include peer-reviewed publications, implementation guidance briefs, and a scalable decision-support framework adaptable to additional counties.

Expected outcomes and impact.

The programme will produce:

(i) the first empirical quantification of ADHDs recognition delays in Kenya,

(ii) probabilistic models linking symptom trajectories to educational outcomes, and

(iii) a validated decision-support tool for early identification and intervention.

By translating evidence into practical guidance, this research will reduce invisible attrition and inform policy on school-based mental health interventions.

Scientific ambition and career trajectory

This programme establishes a methodological niche at the intersection of neurodevelopment, longitudinal inference, and health systems research in sub-Saharan Africa. By integrating Bayesian misclassification, survival analysis, and implementation-informed modelling within school systems, it demonstrates that rigorous, locally co-designed statistical science can generate measurable progress toward health equity. It positions me to lead a sustained programme in African neuro-statistics, embedding advanced quantitative methods within regional mental health research infrastructures.

References

1. American Psychiatric Association. *What is attention-deficit/hyperactivity disorder (ADHD)?* American Psychiatric Association. 2026. Available at: https://www.psychiatry.org/patients-families/adhd/what-is-adhd (Accessed 12 February 2026).
2. World Health Organization. *Mental disorders fact sheet.* World Health Organization. 2022. Available at: https://www.who.int/news-room/fact-sheets/detail/mental-disorders (Accessed 11 February 2026).
3. Polanczyk GV, Salum GA, Sugaya LS, Caye A, Rohde LA. Annual research review: a meta-analysis of the worldwide prevalence of mental disorders in children and adolescents. *Journal of Child Psychology and Psychiatry.* 2015;56(3):345–365.
4. Kenya National Examinations Council (KNEC). *Competency Based Assessment: Age Based (Regular) Framework.* Nairobi: Kenya National Examinations Council; 2021.
5. Ministry of Health, Government of Kenya. *Kenya Mental Health Policy 2015–2030: Towards Attaining the Highest Standard of Mental Health.* Nairobi: Ministry of Health; 2015.
6. Bunifu ERP. *A walk-through Kenya's new (2-6-6-3) education curriculum: subjects to be taught.* 2018. Available at: https://erp.bunifu.co.ke/2018/01/31/a-walk-through-kenyas-new-2-6-6-3-education-curriculum-subjects-to-be-taught/ (Accessed 11 February 2026).
7. Committee to Evaluate the Supplemental Security Income Disability Program for Children with Mental Disorders; Board on the Health of Select Populations; Boat TF, Wu JYT, editors. Clinical characteristics of attention deficit hyperactivity disorder. In: *Mental Disorders and Disabilities Among Low-Income Children.* Washington (DC): National Academies Press; 2015. Chapter 6.
8. Atwoli L, Owiti P, Manguro G, Ndambuki D. Attention deficit hyperactivity disorder symptom self-report among medical students in Eldoret, Kenya. *African Journal of Psychiatry.* 2011;14(4):259–264.
9. Wamithi S, Ochieng R, Njenga F, Akech S, Macharia WM. Cross-sectional survey on prevalence of attention deficit hyperactivity disorder symptoms at a tertiary care health facility in Nairobi. *Child and Adolescent Psychiatry and Mental Health.* 2015;9:1.
10. Kinyanda EK, Kong E, Nakku J, Mukama JBF, Nalugya J, Meffert SM, Mitton VB, Nakigudde J, Mutyaba J, Musisi S. Child and adolescent mental health in sub-Saharan Africa: a narrative review. *International Journal of Mental Health Systems.* 2022;16(1):1–18.
11. Nafi O, Shahin A, Tarawneh A, Samhan Z. Differences in identification of attention deficit hyperactivity disorder in children between teachers and parents. *Eastern*

Mediterranean Health Journal. 2020;26(7):834–838.
12. Magai DN, Malik JA, Koot HM. Emotional and behavioral problems in children and adolescents in central Kenya. *Child Psychiatry & Human Development.* 2018;49(4):659–671.
13. Achenbach TM, Rescorla L. Multi-informant assessments of child and adolescent mental health: a meta-analytic review of cross-informant correlations. *Psychological Bulletin.* 2015;141(4):858–900.
14. Ayano G, Yohannes K, Abraha M. Epidemiology of attention-deficit/hyperactivity disorder (ADHD) in children and adolescents in Africa: a systematic review and meta-analysis. *Annals of General Psychiatry.* 2020;19:21.
15. Langley K, Thapar A, Riglin L, et al. Parent–teacher agreement on hyperactivity and inattention symptoms in a national population sample. *Journal of Child Psychology and Psychiatry.* 2024. doi:10.1111/jcpp.13921.
16. Narad ME, Garner AA, Peugh J, et al. Parent–teacher agreement on ADHD symptoms across development. *European Child and Adolescent Psychiatry.* 2020;29:1601–1612.
17. Rommelse NNJ, et al. Multi-informant assessment in ADHD: clinical implications and contextual interpretation. *Children.* 2023;10(7):914.
18. De Los Reyes A, et al. Advancing multi-informant assessment of child and adolescent mental health. *Child and Adolescent Psychiatry and Mental Health.* 2023;17:56.
19. Quinn PO. Treating adolescent girls and women with ADHD: gender-specific issues. *Journal of Clinical Psychology.* 2005;61(5):579–587.
20. Young S, Adamo N, Ásgeirsdóttir BB, et al. Females with ADHD: an expert consensus statement taking a lifespan approach providing guidance for the identification and treatment of attention-deficit/hyperactivity disorder in girls and women. *BMC Psychiatry.* 2020;20:404.
21. De Los Reyes A, Thomas SA, Goodman KL, Kundey SMA. Principles underlying the use of multiple informants' reports. *Annual Review of Clinical Psychology.* 2013;9:123–149.
22. Achenbach TM, Rescorla LA, Ivanova MY. International epidemiology of child and adolescent psychopathology: multi-informant and multicultural findings. *Comprehensive Psychiatry.* 2016;66:1–8.
23. Novak I, Morgan C, Adde L, et al. Early, accurate diagnosis and early intervention in cerebral palsy: advances in diagnosis and treatment. *JAMA Pediatrics.* 2017;171(9):897–907.
24. Thapar A, Cooper M. Attention deficit hyperactivity disorder. *The Lancet.* 2016;387(10024):1240–1250.
25. Siddique J, Brown CH, Hedeker D, Duan N, Gibbons RD, Miranda J, Lavori PW. Missing data in longitudinal trials – part B, analytic issues. *Psychiatric Annals.*

2008;38(12):793–801.
26. Twisk JWR. Missing data in longitudinal studies. In: *Applied Longitudinal Data Analysis for Medical Science: A Practical Guide.* Cambridge: Cambridge University Press; 2023:201–215.
27. Austin PC, Fine JP. Practical recommendations for reporting Fine–Gray model analyses for competing risk data. *Statistics in Medicine.* 2017;36(27):4391–4400.
28. Collins GS, Reitsma JB, Altman DG, Moons KGM. Transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (TRIPOD): the TRIPOD statement. *Annals of Internal Medicine.* 2015;162(1):55–63.
29. Ministry of Health, Kenya. *Kenya Mental Health Policy 2015–2030.* Nairobi: Ministry of Health; 2015.

Have you used generative AI in the writing of your application?

No - I have not used generative AI

Does your proposal involve human participants, human biological material (including cell lines) or data about living people?

Yes

Does your proposal involve a clinical trial?

No

Details of study design for research involving human participants

Study design

This programme is a prospective, school-based longitudinal cohort study conducted in two Kenyan counties representing urban and peri-urban contexts. Approximately 1,500 adolescents aged 10–14 years will be recruited from 15–25 public primary schools using a multistage sampling strategy. Schools will be selected proportionate to county populations, and learners meeting inclusion criteria will be randomly selected within each school. Cerebral palsy (CP) learners will be included as a visible neurodevelopmental comparator to estimate differences in recognition delays. This approach preserves representativeness, reduces selection bias, and enables estimation of structural disadvantages for less visible conditions such as ADHD.

All participants will be followed longitudinally from baseline through the transition to junior secondary school, with repeated multi-informant assessments every six months over 24 to 60 months. ADHD symptoms and functional impairment will be captured using the Vanderbilt ADHD Diagnostic Rating Scales (parent and teacher versions), complemented by broader emotional and behavioral measures via the Strengths and Difficulties Questionnaire. A clinically assessed validation subsample will undergo structured diagnostic interviews using the K-SADS-PL or MINI-KID to anchor Bayesian misclassification models. Educational outcomes, including grade repetition, school transfer, attendance, and referral documentation, will be extracted from school records. CP comparator data will be collected similarly to allow direct benchmarking of recognition pathways.

Sample size is a hypothetically justified starting cohort of 1,500 learners, sufficient to capture the expected number of events for survival analysis, support robust longitudinal modelling, and accommodate up to 30–40% attrition. Sample size calculations will formally account for event rates, school-level clustering, repeated measures, and the need to detect meaningful differences in time-to-recognition and educational outcomes. Analytical strategies include joint longitudinal–survival models, Bayesian misclassification, and inverse-probability weighting to address informative dropout, measurement uncertainty, and clustering.

All participants will be followed consistently, with clinical assessments for the CP comparator group and the validation subsample conducted at baseline and annually. School transfers will be treated as time-varying covariates, capturing the impact of mobility on symptom trajectories and care pathways. Adolescents identified with urgent mental health needs will be immediately referred through structured pathways. Research

staff will be trained in child protection, confidentiality, and psychological first aid, ensuring participant safety and ethical conduct. Sex assigned at birth and self-reported gender identity will be recorded, with analyses stratified or adjusted to explore differences in experience and outcomes.

This design ensures a rich, ethically grounded dataset that captures both structural delays in recognition and longitudinal outcomes, allowing robust estimation of invisible attrition while providing actionable insights for policy and intervention in low-resource school contexts.

Outline your strategy for recruitment and describe the inclusion or exclusion criteria for study participants (if applicable).

Participants will be recruited through a multistage, probability-based sampling approach to ensure representativeness across urban and peri-urban contexts in the two selected Kenyan counties. Schools will be randomly selected proportional to the county-level population of upper primary (Grades 4–6) learners and JSS(Grade 7-9). Within each participating school, learners meeting inclusion criteria will be randomly invited to participate, preserving unbiased representation and minimizing selection or stigma-related bias.

Inclusion criteria comprise: adolescents aged 10–14 years enrolled in participating upper primary schools, whose parents or caregivers provide informed consent, and who provide their own assent. Cerebral palsy learners within the same schools will be included as a visible neurodevelopmental comparator to enable benchmarking of recognition delays. Exclusion criteria include severe sensory or cognitive impairments preventing meaningful participation in questionnaire-based assessments or repeated follow-up, or inability to provide assent/consent.

All participants will undergo baseline screening using validated parent- and teacher-reported instruments (Vanderbilt ADHD Diagnostic Rating Scales and Strengths and Difficulties Questionnaire). A clinically assessed subsample will receive structured diagnostic interviews (K-SADS-PL or MINI-KID) to validate screening and anchor probabilistic modelling. Educational and referral data will be collected for all participants from school and EARC records.

This strategy ensures inclusion of learners from diverse socioeconomic, geographic, and gender backgrounds, while protecting participant welfare. Sex assigned at birth and self-reported gender identity will be recorded and incorporated in analyses. Randomized school and pupil selection mitigates selection bias, ensures equitable inclusion of underserved populations, and captures the full range of ADHD symptom expression and systemic delay patterns.

How have you involved patients, participants, patient advocacy groups or communities in developing this proposal? What ongoing involvement will they have in the research?

This programme has been developed in consultation with clinicians and child psychiatrists to ensure clinical relevance, feasibility, and ethical integrity. Expert input informed the selection of multi-informant assessment instruments, follow-up timing, and overall study approach. Evidence from longitudinal mental health research in Kenyan schools, including repeated assessments by the African Population and Health Research Center (APHRC), indicates that school-based data collection is feasible, acceptable, supporting confidence in the proposed methods. While formal engagement with school administrators, teachers, county educational officers, and parent representatives has not yet occurred, structured stakeholder involvement will be embedded throughout implementation. Early consultations will refine data collection procedures, ensure contextual appropriateness, and guide strategies to minimize stigma and support retention. Adolescents and caregivers will provide feedback on questionnaires and communications to enhance clarity and acceptability. Ongoing engagement will include school-based advisory panels of teachers, parents, and adolescents to review progress, advise on emerging findings, and tailor the decision-support tool to local needs. Results will be shared with participants, caregivers, and schools via workshops and accessible reports, fostering transparency and co-ownership of knowledge. This approach ensures the research is scientifically rigorous, ethically grounded, and responsive to the communities it serves, while remaining feasible and culturally appropriate.

Describe the oversight arrangements for the study. For example, for a clinical trial. If your research will not have a separate group responsible for oversight you must explain why not.

The study will be overseen by a Study Oversight Committee comprising senior researchers in child and adolescent mental health, biostatistics, and public health, together with representatives from participating schools, local health authorities, and youth/parent advisory panels. This committee will meet biannually to review overall progress, ethical conduct, data quality, and participant safety.

A dedicated Data Management and Monitoring Subgroup will provide continuous oversight of data integrity, confidentiality, and follow-up procedures. This group will also monitor any adverse events or psychological distress arising from assessments and ensure timely response according to pre-defined protocols.

As an observational, longitudinal study rather than a clinical trial, no formal Trial Steering Committee or Data Safety Monitoring Board is required. Nonetheless, the combined oversight structure ensures rigorous monitoring of ethics, participant welfare, and scientific quality throughout the study. By embedding both expert and community perspectives, these arrangements provide accountability, transparency, and adaptive guidance, ensuring that the research is conducted safely, responsibly, and in alignment with local and international ethical standards.

Who has, or will, review the ethics of the project and when? Detail any other regulatory approvals you have, or will try to get.

Before initiating the study, ethical approval will be obtained from KEMRI SERU (Kenya Medical Research Institute Scientific and Ethics Review Unit) and from relevant county-level education and health authorities. Administrative approvals will also be secured from each participating school, and informed parental consent and adolescent assent will be obtained for all participants. All research activities will comply with Kenyan data protection legislation, ensuring secure storage, restricted access, and confidentiality of personal information. Clinical assessments conducted for validation purposes will follow national guidelines for adolescent mental health. The study will also adhere to any additional local regulatory requirements identified during site engagement. The Study Oversight Committee will provide ongoing review of ethics and governance, with formal updates submitted at key project milestones or whenever significant protocol amendments occur. This framework ensures that the research aligns with international standards for human-participant studies while remaining fully responsive to the local legal, social, and cultural context. It balances participant safety, regulatory compliance, and operational feasibility, providing a robust ethical foundation for longitudinal, school-based research in Kenya.

Will you be using facilities, staff or patients within the National Health Service (NHS) in the UK?

No

Which organisation/s have agreed to act as the formal sponsor for your project?

KIPRE (Kenya Institute of Primate Research) will act as the formal sponsor for this project, overseeing ethical approval, governance, and regulatory compliance. Potential collaboration with organizations such as AMREF Health Africa or the African Population and Health Research Center (APHRC) may be explored to support field implementation and community engagement.

Confirm you have, or you will try to get, appropriate informed consent to use any potentially commercially exploitable results from data, tissues or samples derived from human participants.

Not applicable – this research will not generate any data, tissues, or samples with potential commercial applications. All data collected will be used solely for understanding adolescent mental health trajectories and care pathway delays, with findings intended for policy, academic, and methodological purposes.

Describe how you will undertake your proposed research in an environmentally sustainable way.

To conduct this research sustainably, we will adopt multiple strategies to reduce environmental impact throughout the study. Data collection will be largely school-based and digital, minimizing paper use and physical materials. Any printed materials will use recycled paper, and equipment and consumables will be reused wherever safe and feasible. Electronic data management and cloud-based storage will further reduce reliance on physical resources.

We will prioritize virtual meetings, training sessions, and data-sharing platforms to minimize travel for collaborators and research staff. Field visits for participant engagement or follow-up will be combined to reduce trips, and fuel-efficient vehicles will be used when travel is essential.

Environmental impact will be monitored through simple internal metrics, such as tracking travel distance, paper use, and energy consumption in the office and field activities. Staff will be encouraged to adopt sustainable practices, including energy-efficient devices, recycling, and careful management of consumables.

Overall, by leveraging digital tools, minimizing unnecessary travel, and optimizing the use of materials, the project will ensure that research activities are conducted in an environmentally responsible and sustainable manner without compromising scientific quality or participant engagement.

Completed

Outputs management and sharing

Provide an outputs management plan

This research will generate multiple types of outputs, including de-identified longitudinal datasets on adolescent mental health symptoms and care-seeking, analytic code (R scripts and Shiny dashboards), psychometric validation reports for locally adapted screening tools, and methodological frameworks for handling uncertainty, dropout, and clustering in low-resource longitudinal data. These outputs are designed to advance public health by enabling robust evidence on adolescent mental health dynamics and supporting reproducible, policy-relevant analyses in Kenya and similar settings.

Data and metadata management: All datasets will be accompanied by comprehensive metadata describing participant characteristics, variable definitions, data collection procedures, and quality control protocols. Standard operating procedures will document survey administration, clinical validation, and school- and household-level data linkage. Code and workflows will include detailed comments and instructions to ensure reproducibility.

Timing and availability: De-identified datasets and analytic code will be prepared for sharing following completion of primary analyses and publication of key findings, anticipated within six months of final data cleaning. Outputs will be made openly accessible through recognized repositories such as the **African Mental Health Research Data Platform**, the **UK Data Service**, or **Zenodo**, ensuring persistent identifiers (DOIs) and discoverability. Research outputs will also be disseminated through conference presentations, workshops with policymakers and educators, and accessible summaries for local communities.

Access restrictions: Participant confidentiality and ethical approvals dictate that identifiable data will not be shared publicly. Access to individual-level data will be controlled via a formal request process, requiring institutional review and a data-sharing agreement. Aggregated or anonymized datasets will have minimal restrictions and can be freely used by other researchers under a Creative Commons license.

Data storage and preservation: All raw and processed data will be stored on secure servers at KIPRE with routine encrypted backups. Cloud-based mirrors with restricted access will serve as secondary storage. Metadata and documentation will follow FAIR

principles to ensure datasets remain findable, accessible, interoperable, and reusable. Resources for outputs management, including dedicated research staff time and storage infrastructure, are built into the project budget.

Intellectual property (IP) management: This research may generate IP in the form of analytic tools, statistical methods, and software dashboards. These will be protected under KIPRE's institutional IP policy, with licensing to ensure open access for academic and public health use. IP will be used primarily to maximize health benefit, enabling other researchers, policymakers, and implementers to apply validated methods for adolescent mental health surveillance and intervention planning. No third-party ownership or exclusive access arrangements are planned; all outputs will remain non-commercially licensed for research and public health purposes.

Point of contact for IP and outputs management: Dr. Danson Mwangi, PhD, Director Grant management, Capacity building and Partnerships, Kenya Institute of Primate Research, mwangikareri@primateresearch.org , will act as the primary contact for Wellcome regarding IP protection, management, and commercialization discussions.

By combining rigorous metadata standards, open-access policies, secure storage, and FAIR-aligned practices, this plan ensures that all research outputs are maximally discoverable, reusable, and impactful, supporting evidence-based improvements in adolescent mental health care in Kenya and across low-resource settings.

Select the approach you will use to maximise the impact of your significant research outputs to improve health and benefit the wider research community. If an outputs management plan is not needed, select 'Not applicable'.

Make research outputs available for access and re-use

Completed

Collaborations

Are any collaborations essential for this proposal?

Yes

List any key collaborators (name and org). Provide a very brief outline of their role in the proposed research and any key resources they will contribute.

Name	Organisation	Outline of role in proposed research (75 words maximum)
Dr. Allan Ole Kwallah, PhD.	KEMRI(Kenya Medical Research Institute)	Dr. Allan will act as a local scientific mentor, helping ensure the study fits real clinical and community practice in Kenya. Drawing on his experience at KEMRI, he will advise on safe referral pathways, field procedures, and stakeholder engagement. He will also support interpretation of findings for policy relevance and help navigate regulatory processes and partnerships with health and school systems.
Dr. Peter Kuria Waweru	KUTRRH (Kenyatta University Teaching, Referral and Research Hospital)	Dr Waweru will provide clinical oversight for adolescent participants identified as high risk, advising on referral pathways, safeguarding procedures, and integration with existing mental health services. He will support development of clinically appropriate assessment protocols and ensure ethical management of urgent cases such as suicidality. His involvement strengthens the project's connection to real-world care systems and helps translate findings into practical guidance for clinicians and schools within Kenyan health settings.
Mrs. Martha Maina	KEMRI(Kenya Medical Research Institute)	The collaborator will provide technical and analytical support for the project, assisting with data management, cleaning, and preliminary analysis. They will help implement structured research protocols, support data collection tools, and ensure high-quality handling of the project's datasets. Their expertise in data analysis and research methodology will contribute to accurate interpretation and reporting of findings.

I confirm that the collaborators named here have agreed to be involved, as described, in the proposed research and are willing for their details to be included as part of this application.

I confirm

Completed

Location of activity

Will the funded activity take place at more than one location?

No

Wellcome Trust supported facilities

No

Will you need funds to be awarded directly to more than one location?

No

Completed

Research involving animals

Does your proposal involve the use of animals or animal tissue?

No - does not involve animals or animal tissue

Completed

Risks of research misuse

Confirm that you have considered whether your proposed research could generate outcomes that could be misused for harmful purposes.

I confirm

Have you identified any tangible risks of this type?

No

Completed

Freedom to operate and conflicts of interest

Describe any freedom to operate or other intellectual property related issues that might affect your ability to do the proposed research or to use, share or commercialise the research outputs. Explain how you will address these.

Not applicable. This research does not rely on proprietary technologies, patented inventions, or databases controlled by third parties. All study instruments, data collection tools, and analytical methods will be locally validated or developed openly, and outputs will be shared under open access principles as outlined in the Outputs Management Plan. No agreements with commercial or academic organizations will restrict the use, sharing, or commercialization of research outputs.

Describe any conflicts of interest which might affect your ability to do the proposed research or to share or commercialise the research outputs.

Not applicable. Neither I nor any member of the research team hold consultancies, advisory roles, equities, or directorships in organizations that could influence the conduct, outcomes, or sharing of this research. There are no financial or personal interests that might affect the study, and all team members will comply with KIPRE's conflict of interest policies.

Completed

Lead applicant details

Full name

Mr Patrick Mwaura

Current position

Job title	Start date	End date – if known	Department	Organisation	Main employer
Research Scientist	01/04/2019		Data Science and Analytics Section	Kenya Institute of Primate Research	Yes

Previous roles

Job title	Department	Organisation Name	Country	Start date	End date
Founder/ CEO	Owner	Mwas Computer Limited		01/01/2017	31/12/2018
Research Assistant	Population Statistics	Kenya National Bureau of Statistics		01/09/2015	30/08/2017

Have you taken any career breaks?

No

Add your current or most recent salary

Salary grade	Basic annual salary	Currency
NM8	504000	Kenyan Shilling

Add your current and previous qualifications and any relevant training

School	Country	Degree or qualification	Subject	Start date	End date (or expected)
Hasselt University	Belgium	Master of Science (MSc)	Statistics and Data Science (Biostatistics)	01/09/2023	14/07/2025

Are you a healthcare professional?

No

Are you clinically active?

No

Have you completed a PhD, or an equivalent higher research degree?

No

Provide details of your equivalent research experience.

I have not completed a PhD. However, I have over four years of equivalent research experience, including leading and contributing to biomedical and public health research projects since 2019, providing statistical and methodological leadership, developing and applying advanced statistical models, supporting study design and analysis across multiple funded studies, and disseminating research outputs through peer-reviewed publications and international scientific presentations.

Will your host organisation employ you on a part-time basis during this grant?

Yes

What percentage of a working week will your host organisation employ you to work?

30

Will you be 'clinically active' during the grant?

No

Current and recent research funding (including Wellcome grants)

Current and recent research funding

None as lead or co-applicant (last five years).

Fellowship support (2021)

In 2021, I was awarded a fellowship that supported training and methodological research in infectious disease modelling, specifically COVID-19 transmission dynamics using data from the Institute of Primate Research COVID-19 Testing and Research Centre. The fellowship supported analytical work and dissemination, including presentation of findings at a scientific meeting in Accra, Ghana.

This fellowship did not constitute independent project funding and did not support a standalone research programme. There is no scientific or financial overlap between this fellowship-supported work and the proposed research.

What percentage of your time will you spend on research during this grant?

50

What percentage of this research time will you spend on this project?

70

Completed

Your research contributions

How have you contributed to the generation of knowledge?

I have contributed to the generation of knowledge through applied statistical modelling, data science, and biomedical research, focusing on longitudinal, survival, and high-dimensional analyses. My work spans infectious disease surveillance, ecological monitoring, and public health research, with outputs that have informed institutional decisions, policy, and scientific understanding. I have developed data collection systems, reproducible analytical pipelines, and dashboards that translate complex analyses into actionable insights for researchers and decision-makers.

Selected research outputs

1. Akunda, I. K., Kariuki, D. W., Matulis, G., **Mwaura, P.**, et al. (2023). *Antimicrobial resistance patterns and characterisation of emerging beta-lactamase-producing Escherichia coli in camels sampled from Northern Kenya.* Veterinary Medicine and Science. [DOI: 10.1002/VMS3.1090]. *Led data management and analysis, providing evidence of emerging resistance patterns in livestock relevant to One Health surveillance.*
2. Oreng, P., **Mwaura, P.**, Kipkalom, R., et al. (2026). *Dysmenorrhea among undergraduate students: A cross-sectional study at Kenyatta University, Nairobi-Kenya.* East African Medical Journal, 102(11), 8835–8849. [DOI: https://www.ajol.info/index.php/eamj/article/view/313621]. *Designed data collection tools and conducted statistical analyses, providing insight into menstrual health in Kenyan university populations.*
3. Fellowship project, CEMA & Cambridge-Africa (2021): *Unraveling Covid-19 Transmission Dynamics in Kenya using a Gaussian Stochastic Process model. Developed and applied stochastic modelling approaches to national COVID-19 testing data, contributing to understanding transmission dynamics.*
4. USAID-PEER One Health Project (2021–2022). *Led data collection and analysis integrating human, animal, and environmental health data, enabling evidence-based One Health interventions.*
5. IPR Nutrition-Ecology of Hirola Antelopes (2021–2024). *Programmed ODK tools and conducted data analysis on longitudinal ecological data, contributing to conservation planning for a critically endangered species.*

6. COVID-19 Testing and Research Centre, KIPRE (2019–2023). *Developed automated data capture and analysis pipelines, supporting laboratory surveillance and real-time reporting for policy decisions.*
7. Natural Products Industries (NPI) (2021–Present). *Led digitization of ethnobotanical data collection and analysis, ensuring compliance with TK&CE Act 2016 and supporting research reproducibility.*
8. **Master's thesis, University of Hasselt (2025):** *"Sample size determination of experiments for standardized performances of solar cooking appliances".* Developed simulation-based approaches for optimal sample size determination in complex experimental designs. Built an R Shiny app to allow researchers to visualise and compare sample size requirements under different scenarios. Contributed to improving design efficiency and reproducibility in experimental studies.

Across these projects, I have applied statistical inference, predictive modelling, machine learning, and longitudinal analysis techniques, ensuring outputs are reproducible, well-documented, and available to collaborators and stakeholders. My contributions have consistently bridged technical expertise and policy-relevant evidence, strengthening institutional research capacity and promoting applied data science in biomedical and ecological contexts.

How have you contributed to the wider research community?

I actively contribute to the wider research community by mentoring, teaching, and fostering collaboration. At KIPRE, I guide early-career researchers and students in statistical modelling, longitudinal and survival analysis, and designing reproducible analytical workflows. I have trained teams in R, Python, REDCap, CSPro, ODK, and SQL, helping build capacity across biomedical, ecological, and One Health research projects.

Beyond training, I engage in interdisciplinary knowledge sharing through workshops and fellowships, including the Africa One Health Network and CEMA, exchanging insights on modelling techniques and research design. I support the planning and analysis of institutional projects, ensuring methodological rigor and reproducibility.

I have contributed to open-access tools, such as an R Shiny app for sample size simulation, and to peer-reviewed publications on antimicrobial resistance and adolescent reproductive health, helping make findings widely accessible.

Through these efforts, I aim not only to strengthen research capacity but also to translate data-driven insights into practical solutions that improve health outcomes and inform policy.

Completed

Your research environment details

What considerations led you to choose this research environment and any sponsors for your research? How will this setting help you to develop your research identity and explore your research questions, and be ready to lead your own independent research programme at the end of the grant?

I selected KIPRE, under the Ministry of Health's Department of Public Health and Professional Standards, because it provides access to rich biomedical and public health data, robust analytical infrastructure, and a collaborative scientific community. Working at KIPRE aligns my research with national health priorities, while its mentorship and interdisciplinary environment allow me to strengthen my methodological skills, develop my research identity, and gain the experience required to lead independent, impactful research in the future.

How will you contribute to a positive and inclusive research culture?

I will foster a positive and inclusive research culture by promoting collaboration across disciplines and encouraging open knowledge sharing within and beyond the team. I plan to support colleagues through mentorship, guidance on best research practices, and structured training in research integrity. I will actively champion diversity and inclusion by creating opportunities for underrepresented groups, ensuring equitable participation in projects, and modelling respectful, transparent communication. By embedding these practices in daily research activities, I aim to build a supportive environment that encourages innovation, learning, and the ethical conduct of research.

Your organisation will give you and the staff employed on your grant a minimum of 10 days a year to do training and continuing professional development. Explain how you will use this time.

I will use the allocated CPD time to strengthen both my technical and leadership skills. This includes attending training on research integrity, ethical data management, advanced statistical methods, and emerging data science tools. I will also participate in workshops on research leadership, people management, and fostering diversity and inclusion, ensuring I can create a supportive and equitable research environment. By combining technical learning with professional development, I aim to enhance my ability to lead projects effectively, mentor colleagues, and maintain a healthy, responsible, and inclusive research culture.

Completed

Your sponsor

Full name

Dr Atunga Nyachieo

Current position

Job title	Start date	End date – if known	Department	Organisation	Main employer
Senior Research Scientist & Director of Research and Product Development	01/04/2010		Research	Kenya Institute of Primate Research	Yes

What support will you provide to the applicant?

As Director of Research and Product Development at KIPRE and Patrick Mwaura's supervisor, I am fully committed to supporting his research on adolescent mental health in Kenya. I will provide the institutional resources he needs to lead this longitudinal study independently, including protected research time, a dedicated workspace, and access to KIPRE's research management office for financial and reporting guidance. I will mentor Patrick in stakeholder engagement, facilitating connections with the Ministry of Health to ensure the project informs public health policy. He will also participate in senior-level strategic planning at KIPRE, strengthening his leadership, project management, and team oversight skills. This study requires advanced longitudinal data handling. I will ensure Patrick gains training in trajectory modeling and other analytical techniques through our institutional partnerships. Finally, I will support the dissemination of findings across East Africa, nurturing Patrick's growth as an independent researcher and positioning him as a regional leader in adolescent neurodevelopmental research.

(For the applicant) Are you intending to work for more than three months outside your administering organisation?

No

Completed

Your mentor

Name, including title (for example Professor, Dr)

Professor. Mwangi Thumbi

Department

CEMA (Center for Epidemiological Modelling and Analysis)

Organisation

University of Nairobi

Title of current post

Co-Director CEMA

Completed

Currency requested

Select the currency you would like the grant to be paid in

EUR

Is this your local currency?

No

What is your local currency?

KES

Explain why you are requesting costs in the selected currency and what exchange rate you have used.

The grant is requested in EUR to manage material foreign-exchange risk and protect the real value of the research budget over the award period. The Kenyan Shilling is subject to volatility and depreciation risk, which could materially impair purchasing power and contractual commitments if costs were denominated locally. Costing EUR provides a stable unit of account for multi-year financial planning, reduces exposure to adverse FX movements, and supports accurate pricing of internationally denominated inputs, including digital infrastructure and specialist services. An exchange rate of 1 EUR = KES 160 was applied conservatively. Foreign-currency management will follow institutional controls and Wellcome policy.

Completed

Costs requested and justification

Are you asking for a lead applicant salary?

Yes

Lead applicant salary

Cost type	Role	Name	Basic annual starting salary	Salary grade or scale	Period on project (months)	% time	Cost requested
Salary	Lead Applicant	Patrick Waweru Mwaura	€22,872.00	Lead applicant	60	30	€6,862.00
							Total: €6,862.00

Are you requesting staff?

No

Are you requesting adjustment support?

No

Are you requesting a relocation allowance?

No

Are you requesting training and continuing professional development?

Yes

Training and continuing professional development

Cost type	Description	Cost requested
Continuing professional development and professional skills training	Targeted CPD in leadership, project management, responsible conduct of research, inclusive research practices, and translational methods, to support supervision and fostering a positive research culture. Specific workshops or conferences will be identified during the project.	€3,125.00
Research skills training	Targeted training in global mental health research methods, including longitudinal mental health assessment, psychometrics, measurement validation, clinical assessment of adolescent mental health, and the integration of mental health data with analytic frameworks. Specific workshops will be identified as the project progresses.	€3,125.00
		Total: €6,250.00

Justification for training and continuing professional development

The project combines advanced statistical methods with adolescent mental health research in Kenyan schools, a context where most affected young people remain undiagnosed. While I have a strong foundation in longitudinal and survival analysis, addressing this work requires additional skills in mental health assessment, school-based intervention research, and contextually sensitive study design. Training and professional development will therefore focus on enhancing understanding of adolescent mental health, early identification, and ethically responsible research with minors.

Research skills training will include psychometric validation, repeated measures assessment, clinical evaluation of high-risk adolescents, and integration of complex mental health data with advanced statistical frameworks, including frailty, marginal, conditional, multistate, and Bayesian models. These activities will strengthen my capacity to produce rigorous, reproducible analyses that are both scientifically robust and policy-relevant.

Continuing professional development in leadership, research integrity, and collaborative skills will ensure effective management of school-based cohorts, stakeholder engagement, and dissemination of findings. Collectively, these opportunities will allow me to translate methodological expertise into actionable insights, establish a distinct research identity, and position me to lead independent research programme tackling adolescent mental health inequities in sub-Saharan Africa.

Are you requesting materials and consumables?

Yes

Materials and consumables

Description	Cost requested
Miscellaneous / redundancy	€7,250.00
Printing and consumables	€3,125.00
Field travel support	€72,500.00
Psychometric testing kits or paper-based backup forms	€6,250.00

Description	Cost requested
REDCap & secure cloud hosting	€15,625.00
Internet / mobile data	€3,125.00
	Total: €107,875.00

Justification for materials and consumables

The longitudinal, school-based nature of this study, spanning 15–25 schools and approximately 1,500 adolescents, requires robust and reliable data collection and management processes. Systems for repeated assessments across multiple schools, secure storage, and centralized data management are critical for ensuring timely, accurate, and high-quality data capture over the five-year project.

Support for field operations, including travel between study sites and procedures to maintain participant safety and ethical compliance, is essential for smooth implementation. Measures for data security, confidentiality, and quality assurance help protect sensitive participant information and ensure that all procedures adhere to ethical standards.

Collectively, these processes and safeguards are essential to ensure the integrity, reliability, and ethical conduct of the study. They enable high-quality longitudinal data collection, secure management, and robust analysis while providing the flexibility needed to adapt to real-world challenges in diverse educational settings.

Are you requesting animals?

No

Are you requesting equipment?

Yes

Equipment costs

Type	Type of equipment	No. of items	Cost per item	Cost of maintenance contract	Cost requested
Equipment purchase	Tablets for data collection	12	€313.00	€31.00	€4,000.00
Computer equipment	Laptops for data management & dashboard development	5	€1,563.00	€313.00	€8,125.00
Equipment purchase	Audio recorders & transcription tools	12	€313.00	€0.00	€500.00
Equipment maintenance	Data security and encryption tools	2	€4,688.00	€0.00	€9,375.00
					Total: €22,000.00

Justification for equipment

The project requests modest capital equipment and essential consumables required for longitudinal data collection, secure data management, and qualitative research. This includes tablets for school-based electronic data capture, laptops for central data management and analysis, audio recorders for qualitative interviews and focus groups, encrypted storage devices for secure data backup, and psychometric kits. Additional consumables such as printing materials, consent forms, assessment manuals, and questionnaires will support continuity in fieldwork, while internet and mobile data packages enable smooth, multi-wave uploads from urban, peri-urban, and rural locations. Field travel support ensures staff can transport materials efficiently, and personal protective equipment (masks, sanitizers) safeguards participants and research staff. Backup systems and procedures are in place to mitigate risks of data loss or device failure.

These items are required to support repeated assessments across multiple schools and counties over several years, reduce reliance on paper-based tools, and ensure data quality, security, and timely monitoring. All equipment and consumables will be procured through the host institution's approved processes, ensuring value for money, competitive pricing, and compliance with institutional and Wellcome policies. Devices will be selected based on durability, energy efficiency, and compatibility with REDCap and statistical software, and will be shared across field teams to maximize utilization. All items will be covered by manufacturer warranties for at least the first year, with routine maintenance absorbed within institutional IT support where feasible. No single item exceeds the £100,000 threshold.

Are you requesting a piece of equipment with a list price of £100,000 or more?

No

Are you requesting access charges?

No

Are you requesting overheads?

Yes

Overheads

Description	Cost requested
12.5% of the total grant value	€47,544.00
	Total: €47,544.00

Justification for overheads

Overheads are requested at KIPRE's standard institutional rate of 12.5%, in line with organisational policy and Wellcome guidance. These costs support essential institutional services, including financial management, grants administration, compliance, audit, and governance, which are necessary for the effective delivery and accountability of the proposed research.

Upload a letter from the Finance Director of each organisation. If there is more than one letter, upload these as a single PDF.

- **File name** Draft_ Overheads Mr. Patrick.pdf
- **File size** 249.4KB

The uploaded file is included in this PDF starting on the next page.

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W.H.O. COLLABORATING CENTRE
Translating Life Science
MINISTRY OF HEALTH

16th February 2026
The Wellcome Trust Early Career Grant Committee,
Wellcome Trust, Gibbs Building, 215 Euston Road, London NW1 2BE

Ref No. 345293/Z/26/Z Institutional Overhead Costs

Reference is made to the above: The Kenya Institute of Primate Research (KIPRE), an accredited biomedical research and conservation facility, is a State Corporation under the Ministry of Health, State Department of Public Health & Professional Standards in Kenya.

In accordance with its research mandate, the institute has put in place measures to support and encourage its research scientists to develop independent research programs.

To facilitate the research projects, the institute affords technical, administrative, financial, and grant management services. In addition, KIPRE meets auxiliary costs such as internet access, power and lighting charges, cleaning aids, security and other miscellaneous expenditures.

A sum equivalent to 12.5% of the total project budget (grant value) will be allocated to cover these services throughout the duration of the project. This would be in line with the institutional policy, which is guided by government directive to charge overhead fees from all funded projects.

This letter therefore certifies these overhead costs as a true representative of cost incurred.

Yours Sincerely,

Esther Mukami Njogu
Principal Accountant- Kenya Institute of Primate Research

Are you based at a UK university and requesting overheads on subcontracted costs?

No

Are you requesting travel and subsistence?

Yes

Travel and subsistence costs

Type	Description	How much carbon will this offset (in tonnes)?	Cost requested
Conference attendance	Attendance at regional and international scientific conferences to disseminate findings, receive methodological feedback, and build collaborations in adolescent mental health and biostatistics. Includes economy travel, registration, accommodation, and subsistence for the lead applicant, approximately one conference per year over five years.	10	€10,000.00
Other travel	Domestic travel for longitudinal fieldwork across 15–25 public primary schools in urban, peri-urban, and rural counties. Includes transport, accommodation, and logistics for baseline and repeated follow-up assessments, clinical validation visits, qualitative data collection, and stakeholder engagement meetings with county health and education partners.	36	€19,000.00
Carbon offset	Purchase of accredited carbon credits to mitigate emissions generated from international conference travel and repeated domestic fieldwork travel. Emissions will be reviewed periodically and offset through recognised carbon mitigation schemes.	46	€1,875.00
Subsistence costs	Field subsistence allowances for research staff during extended data collection periods, including meals and incidental costs during overnight stays at study sites where daily	0	€9,000.00

Type	Description	How much carbon will this offset (in tonnes)?	Cost requested
	commuting is not feasible.		
			Total: €39,875.00

Justification for travel and subsistence costs

Travel is essential to the delivery of this longitudinal, school-based study, which spans multiple counties and requires repeated in-person engagement with schools, adolescents, caregivers, and local stakeholders over several years. Domestic travel supports baseline recruitment, follow-up data collection waves, clinical validation assessments, and qualitative interviews, all of which must be conducted within school and community settings to ensure data quality, ethical safeguarding, and participant retention. Given the geographic spread of participating schools across urban, peri-urban, and rural areas, travel and subsistence costs reflect realistic field conditions rather than discretionary activity.

Conference attendance is included to support dissemination of findings and methodological advances in global mental health and biostatistics, and to strengthen collaborations that will enhance the scientific impact and sustainability of the programme. Attendance will be prioritised for meetings with clear relevance to adolescent mental health, longitudinal analysis, and global health, and travel will be undertaken using economy options wherever feasible.

In line with Wellcome's environmental sustainability policy, travel has been carefully planned to minimise unnecessary movement, with route optimisation and clustering of field activities. Carbon emissions associated with essential travel will be mitigated through the purchase of accredited carbon offsets.

Are you requesting overseas allowances?

No

Are you requesting fieldwork expenses?

Yes

Fieldwork expenses

Description	Cost requested
Fieldwork delivery costs for a prospective, school-based longitudinal cohort, including transport to and between schools across urban, peri-urban, and rural counties; per diems and subsistence for field teams during repeated survey waves; community and school engagement meetings; qualitative interviews and focus groups with adolescents, caregivers, and teachers; follow-up and tracing activities for mobile pupils; and coordination and supervision visits required to maintain data quality and participant safety over multiple waves.	€32,000.00
	Total: €32,000.00

Justification for fieldwork expenses

Fieldwork expenses are essential to deliver a large, school-based longitudinal cohort across diverse Kenyan settings. The study involves repeated data collection waves over multiple years in 15–25 public primary schools spanning urban, peri-urban, and rural counties. These costs cover transport of field teams and equipment between schools, per diems and subsistence during data collection, and logistical support for maintaining follow-up in a mobile school-aged population.

Fieldwork activities include quantitative surveys with approximately 1,500 adolescents, clinical validation assessments among high-risk participants, and qualitative interviews and focus groups with adolescents, caregivers, and teachers. Additional resources are required for school and community engagement meetings, participant tracing when pupils transfer schools, and supervision visits to ensure data quality, ethical compliance, and participant safety.

The requested amount reflects a realistic and cost-efficient estimate based on local transport costs, field team size, and the repeated nature of data collection, and is proportionate to the scale and duration of the proposed research.

Are you requesting clinical research?

No

Are you requesting public engagement and patient involvement?

Yes

Public engagement and patient involvement costs

Description	Cost requested
Participant and caregiver engagement workshops	€31,250.00
Co-design and dissemination materials	€6,250.00
	Total: €37,500.00

Justification for public engagement and patient involvement

Meaningful public engagement and patient involvement are central to this project, which focuses on adolescent mental health in school settings. Adolescents, caregivers, and teachers are not only participants but key stakeholders whose perspectives are essential for ensuring that study tools, procedures, and outputs are ethically appropriate, culturally relevant, and practically usable.

The requested costs will support structured engagement workshops at different stages of the project, including early consultations to refine data collection tools, mid-study feedback sessions to reflect on implementation challenges, and dissemination meetings to share findings in accessible formats. These engagements will be conducted in schools and community settings and will prioritise safeguarding, inclusivity, and respectful participation.

Additional resources are required to develop age-appropriate and culturally sensitive materials, including simplified summaries, visual aids, and translated briefs, to enable meaningful dialogue with adolescents and caregivers. Modest transport reimbursements and refreshments are necessary to reduce participation barriers, particularly for caregivers and participants from resource-constrained settings.

These costs are proportionate to the scale and duration of the study and ensure that public engagement is embedded throughout the research lifecycle rather than treated as a one-off activity. By investing in sustained involvement, the project strengthens ethical conduct, improves data quality, and enhances the relevance and impact of findings for policy and practice in Kenya.

Are you requesting other costs?

Yes

Other costs

Type	Description	Cost requested
Specialist publications	Open-access publication fees for peer-reviewed journals and policy briefs to ensure global and regional accessibility of findings, including journals focusing on global mental health and adolescent	€2,813.00

Type	Description	Cost requested
	health.	
Consultancy fees	Sessional professional fees for psychiatrists and senior clinical psychologists providing time-limited clinical oversight, risk assessment, referral guidance, and supervision related to participant safeguarding during longitudinal mental health assessments. Services will be provided on an on-call or per-session basis and do not constitute salaried employment.	€32,500.00
Subject/volunteer expenses	Participant compensation to cover time and inconvenience for adolescents, caregivers, and teachers participating in assessments, interviews, and focus groups across multiple waves, provided in line with ethical approvals.	€12,500.00
Health-related feedback	Costs associated with providing appropriate feedback to participants and caregivers, including referral information, counselling summaries, and follow-up communication for participants identified as at risk during assessments.	€3,750.00
Outputs management plan	Data curation, long-term storage, anonymisation, documentation, and preparation of datasets and metadata for controlled access repositories in line with Wellcome open research and data sharing policies.	€6,250.00
Material for clinical, epidemiological and qualitative research studies	Assessment manuals, screening instruments, qualitative interview guides, transcription templates, and validated psychometric tools required for longitudinal clinical and epidemiological data collection.	€12,500.00
Recruitment, advertising and interviewee travel costs	Community sensitization meetings, recruitment materials, communication costs, and reimbursement of local travel expenses for interviewees attending central assessment or feedback sessions.	€12,500.00
Project-dedicated vehicles	Not requested. Field mobility will be covered through travel and fieldwork expense lines to avoid duplication.	€0.00
PPE	Personal protective equipment for field teams conducting school-based and household assessments, including masks, sanitizers, and basic	€4,688.00

Type	Description	Cost requested
	infection-prevention supplies.	
Computing	Licenses and subscriptions for secure data analysis, qualitative coding, encrypted communication, and dashboard deployment tools not covered under institutional agreements.	€3,125.00
Conference/seminar hosting	Organization of stakeholder dissemination workshops, policy dialogues, and end-of-project scientific meetings involving schools, county health officials, and national stakeholders.	€6,250.00
Costs of ethics review	Ethical review, amendment, renewal, and monitoring fees required by national and institutional ethics review committees for a multi-year adolescent mental health study.	€3,125.00
Clinical Trials Insurance	Not requested. The study is observational and non-interventional, and therefore does not require clinical trials insurance.	€0.00
		Total: €100,001.00

Justification for other costs

The other costs requested are essential to the ethical delivery, scientific quality, and societal impact of this longitudinal adolescent mental health study. Consultancy fees are required to engage psychiatrists and senior clinical psychologists on a sessional basis to provide specialist clinical oversight, participant risk assessment, and referral guidance for adolescents identified with significant mental health concerns. Subject and volunteer expenses are included to compensate participants, caregivers, and teachers for their time and inconvenience across repeated assessment waves, in line with ethical guidance. Health-related feedback costs support responsible communication of findings and referrals to participants and families, particularly where elevated risk is identified.

Costs for recruitment, advertising, and interviewee travel enable equitable participation across diverse school settings, while materials for clinical, epidemiological, and qualitative research support the use of validated tools and high-quality data collection. Computing and outputs management costs ensure secure data handling, long-term storage, and compliance with Wellcome's open research and data-sharing policies. Conference and seminar hosting supports dissemination to policymakers, educators, and health stakeholders, maximizing research uptake. Ethics review and PPE costs are necessary to meet regulatory and safety requirements. No project-dedicated vehicles or clinical trials insurance are requested to avoid duplication and ensure cost efficiency.

Summary of costs requested

Total funding requested in this application

€399,907.00

Completed

Full economic costing

Is your organisation based in the UK?

No